QHM5703: Principles of Machine Learning
Decision Trees

Independent Study

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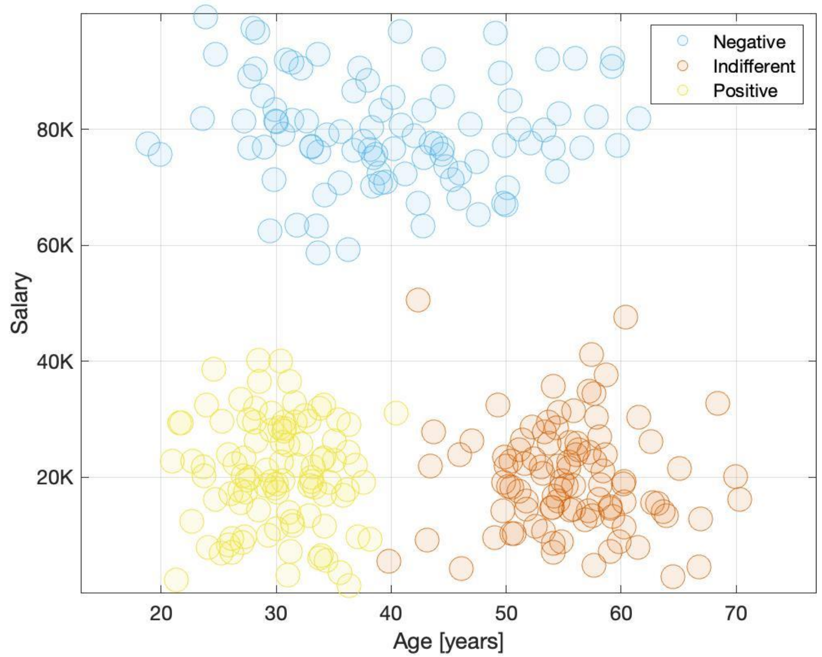
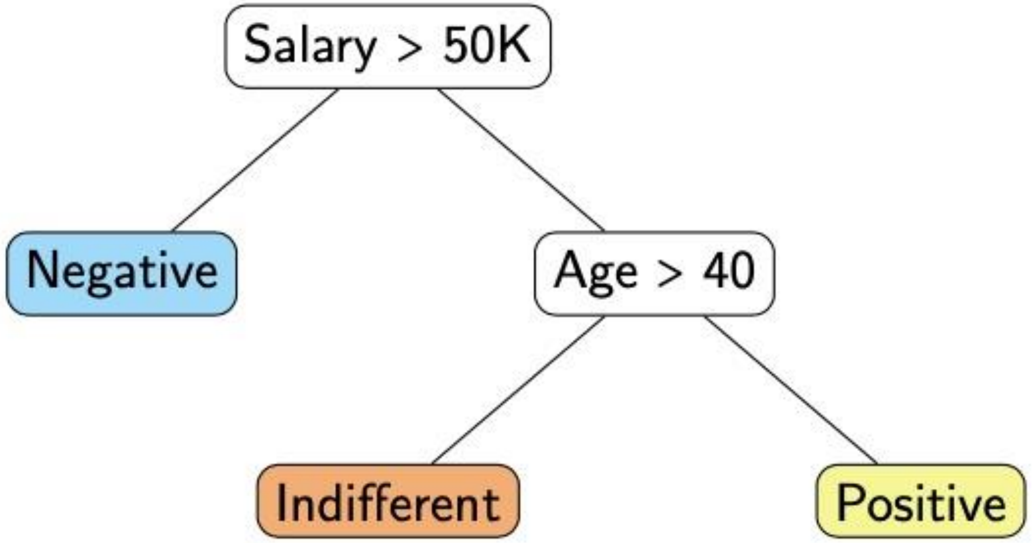
Decision Trees

- **Independent Study**
- Chapter 4: Principles of Machine Learning

  https://pmlbook.github.io/Ch_classification1.html#decision-trees





Decision Trees: Building Trees

In a decision tree, the root corresponds to the whole, unpartitioned dataset and a leaf is one of the decision regions.

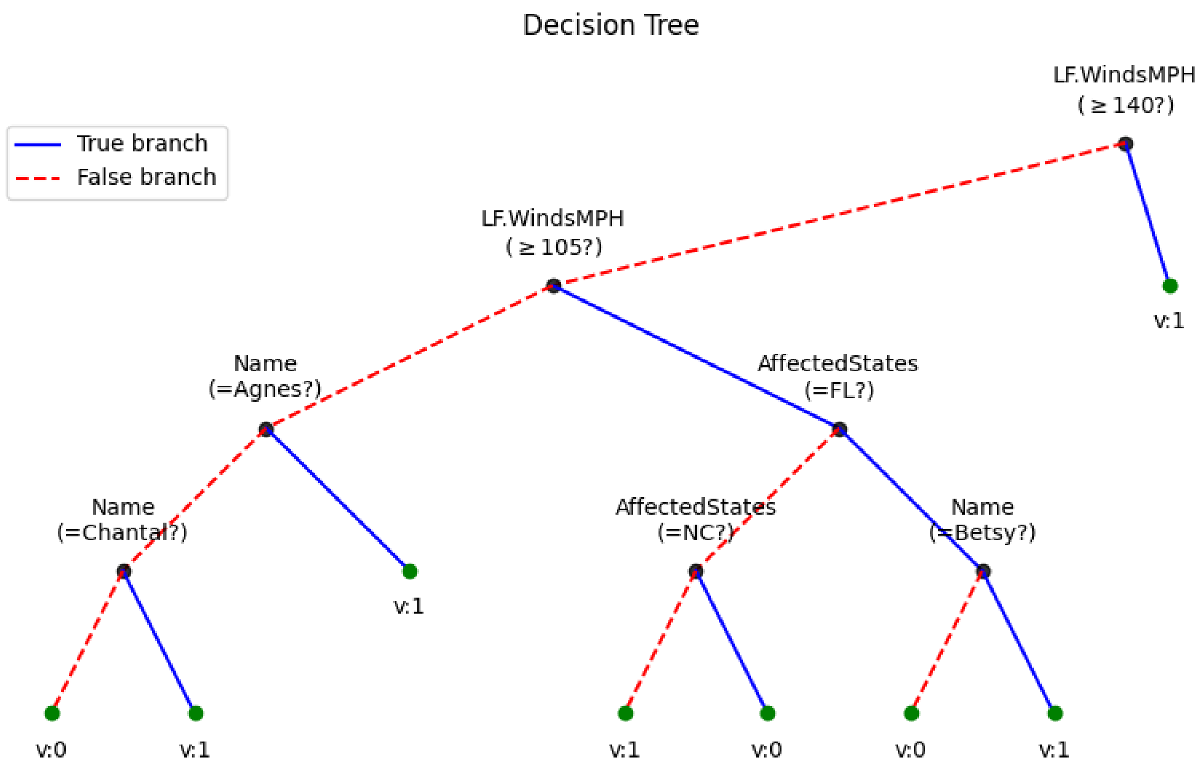
- The goal is to create pure leaves, i.e. containing as many samples from the same class as possible.
- During classification, a sample is assigned to the majority class in the leaf where the sample is located.

Decision trees are built recursively:

- Starting from the root, we recursively split each region into two. Splits are axis-parallel (decisions using one predictor).
- The chosen split is such that the purity of the resulting regions is higher than any other split.
- We stop when a given criterion is met, such as the number of samples in a region.

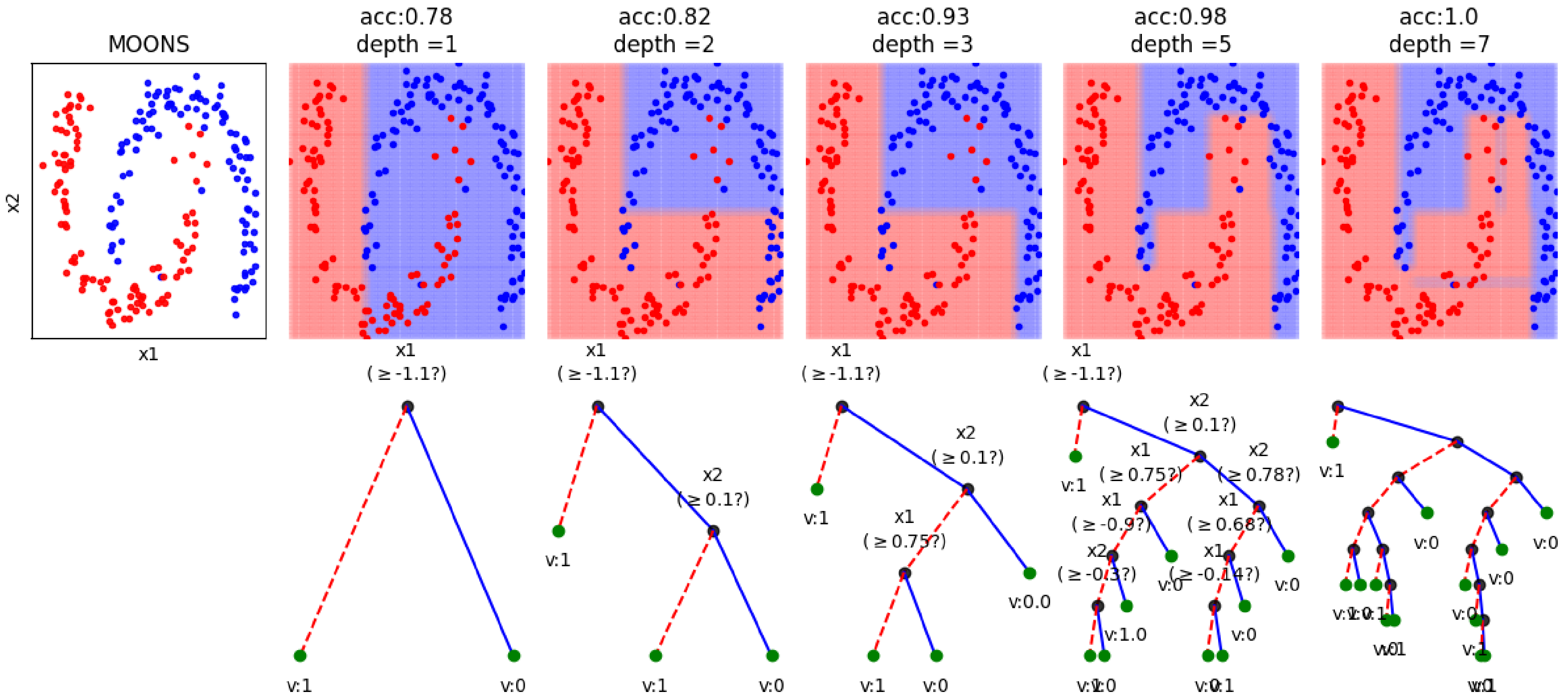
Decision Trees

- Spkit: Examples: ML: Decision Trees
  https://spkit.github.io/auto_examples/index.html#machine-learning



Decision Trees

- Example: Increasing the ***depth*** of tree, more flexibility and overfitting



Decision Trees

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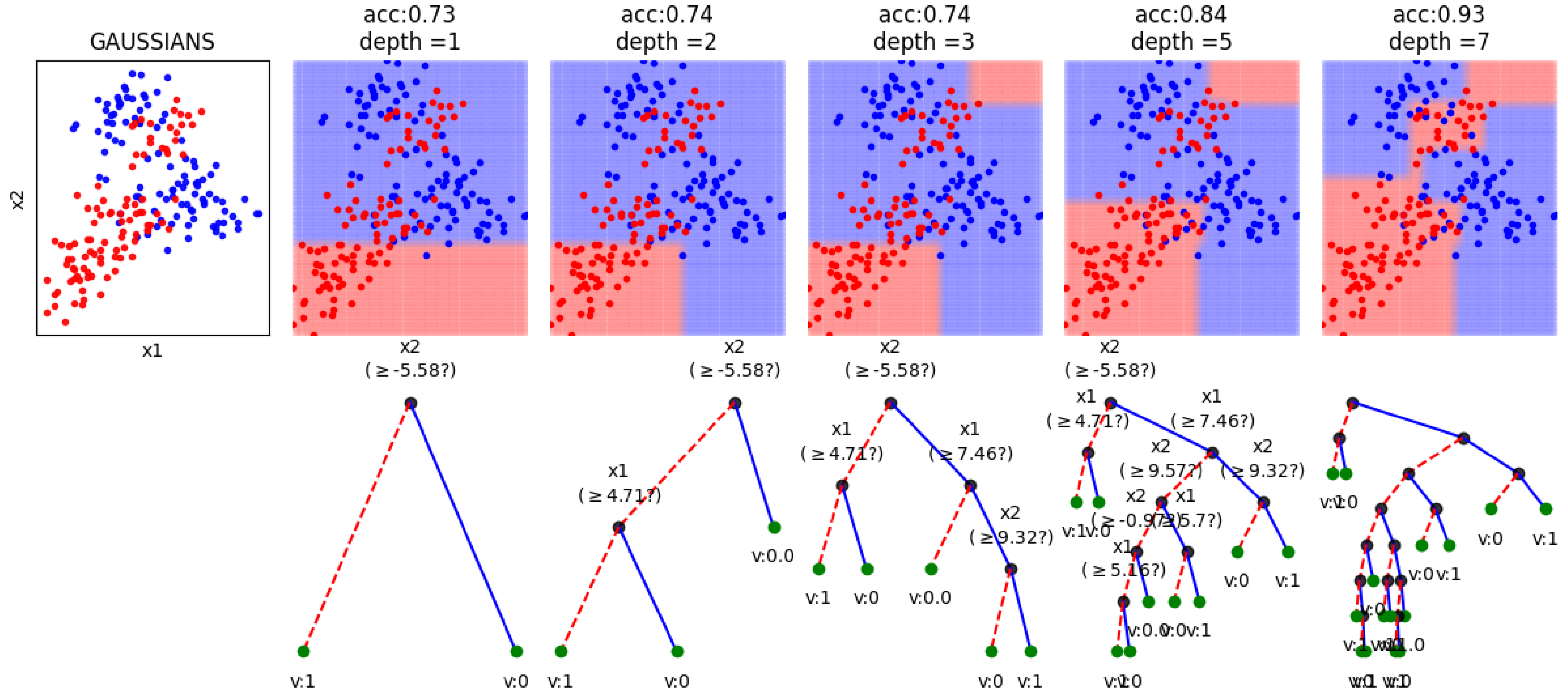

Decision Trees

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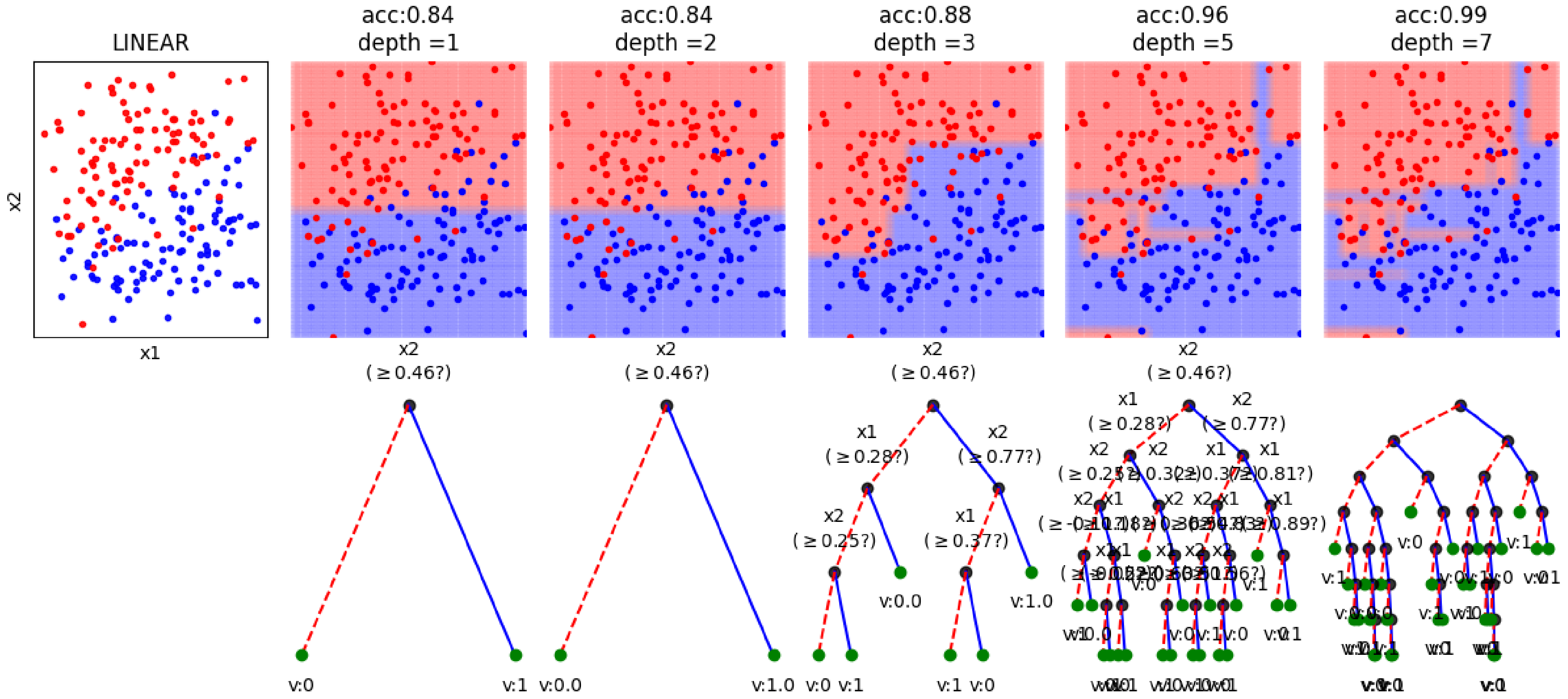

Decision Trees

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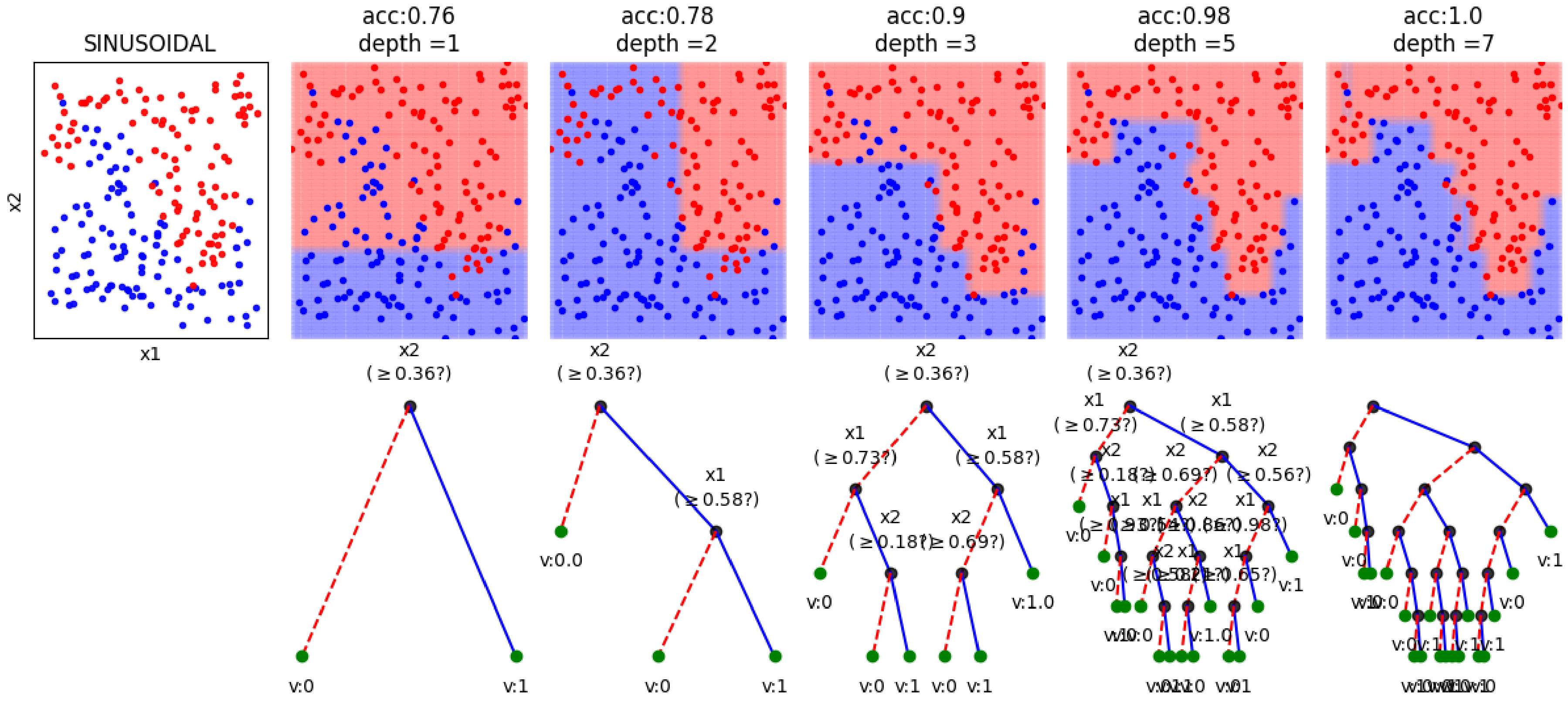

Decision Trees

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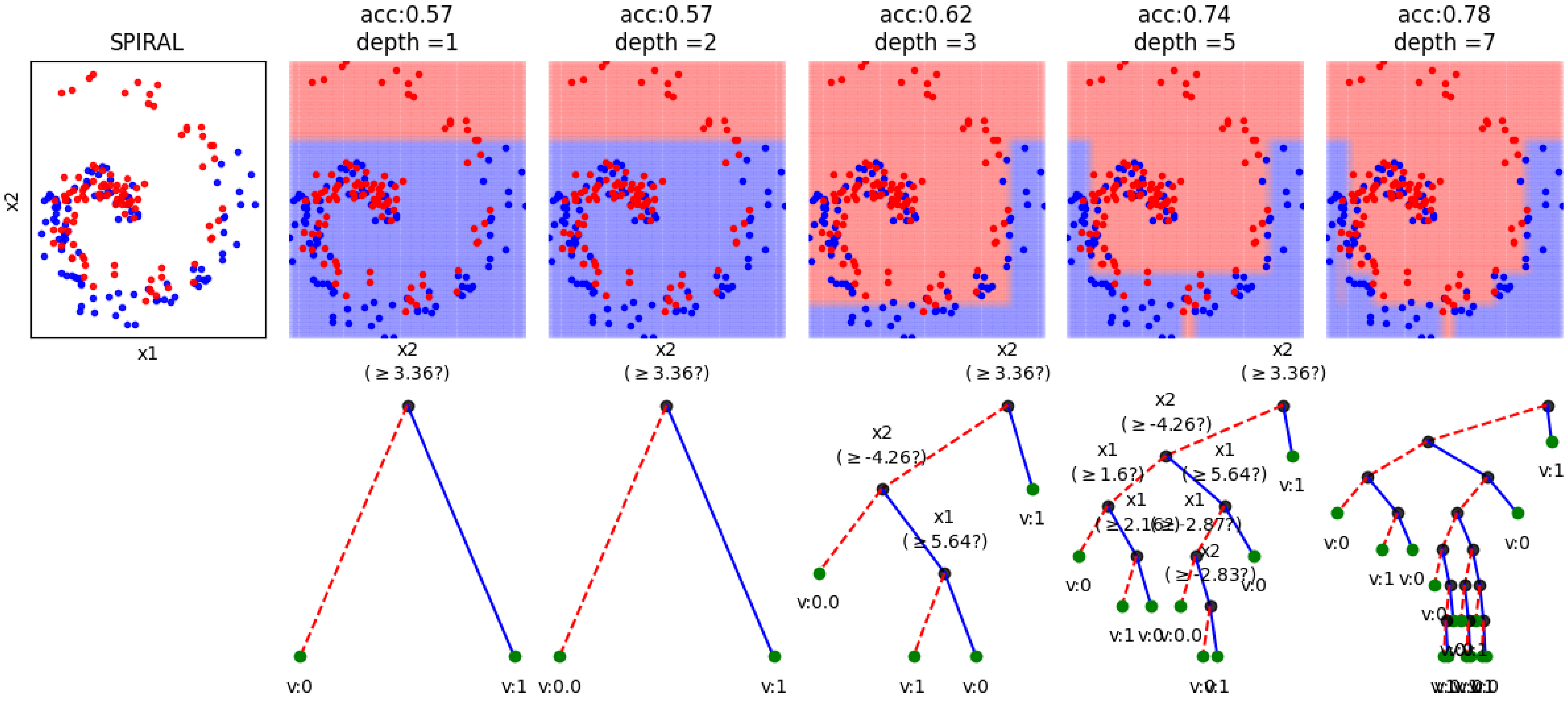
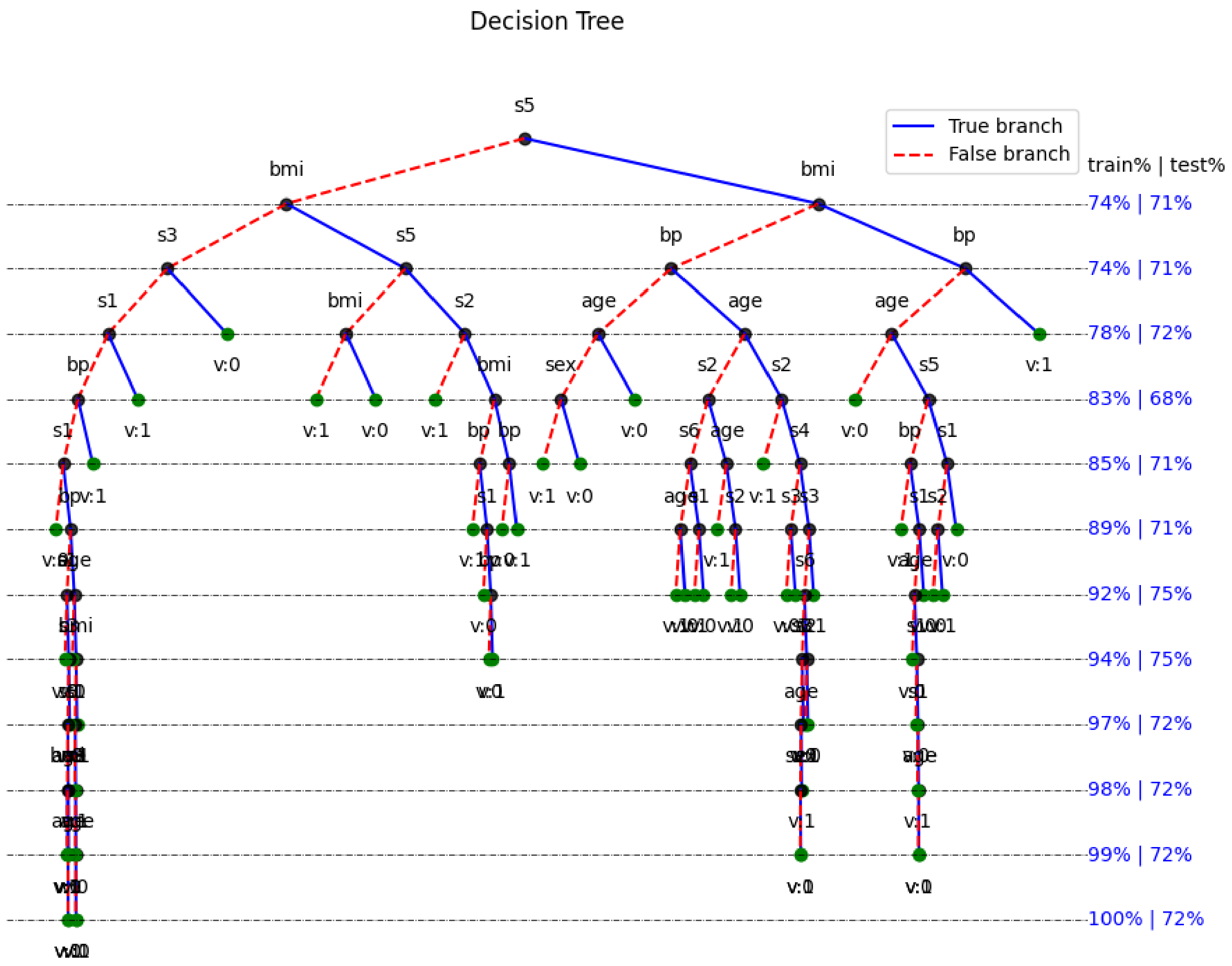


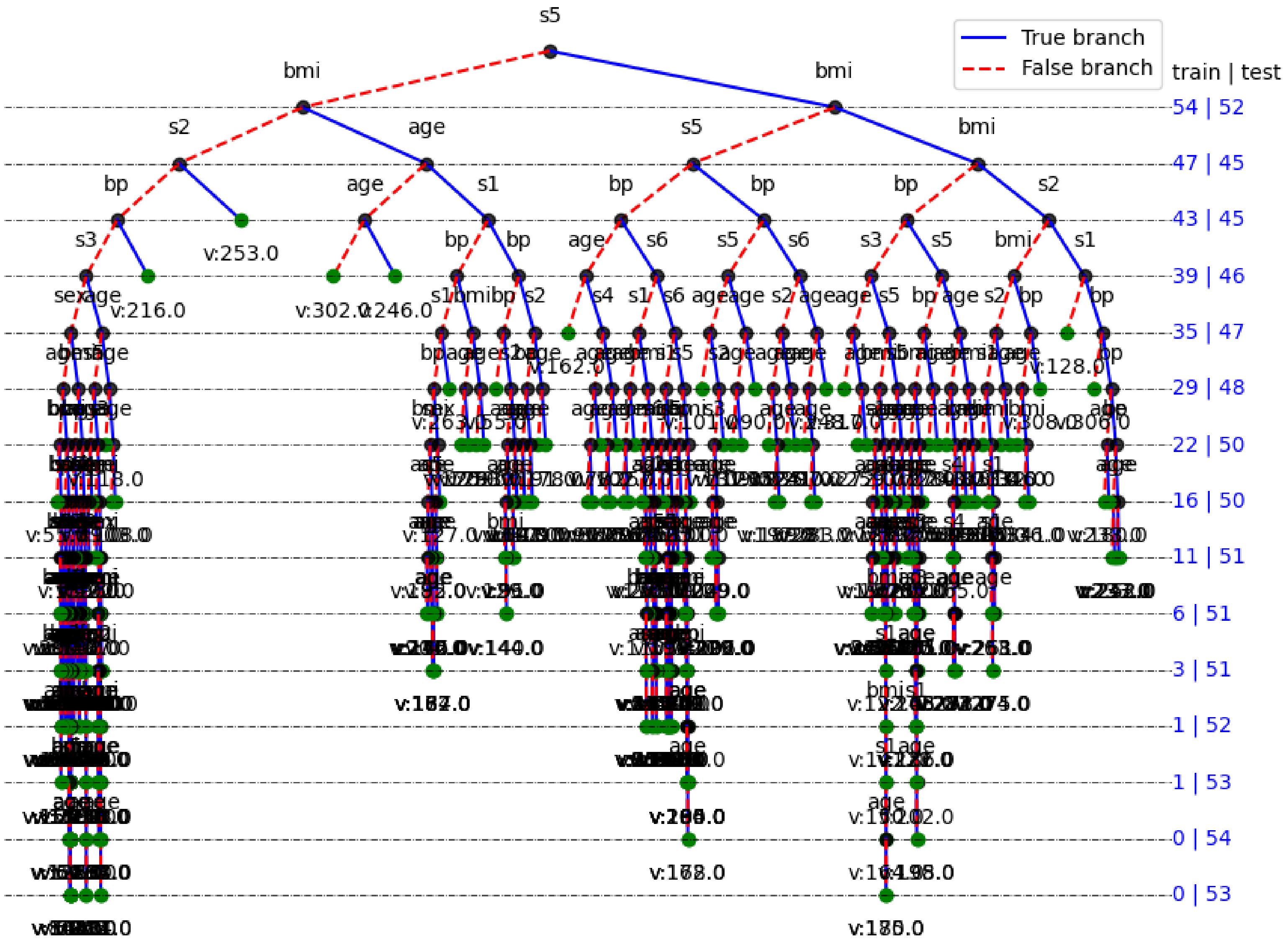
Decision Trees: Classification

- Example: Increasing the ***depth*** of tree, more flexibility and ***overfitting***



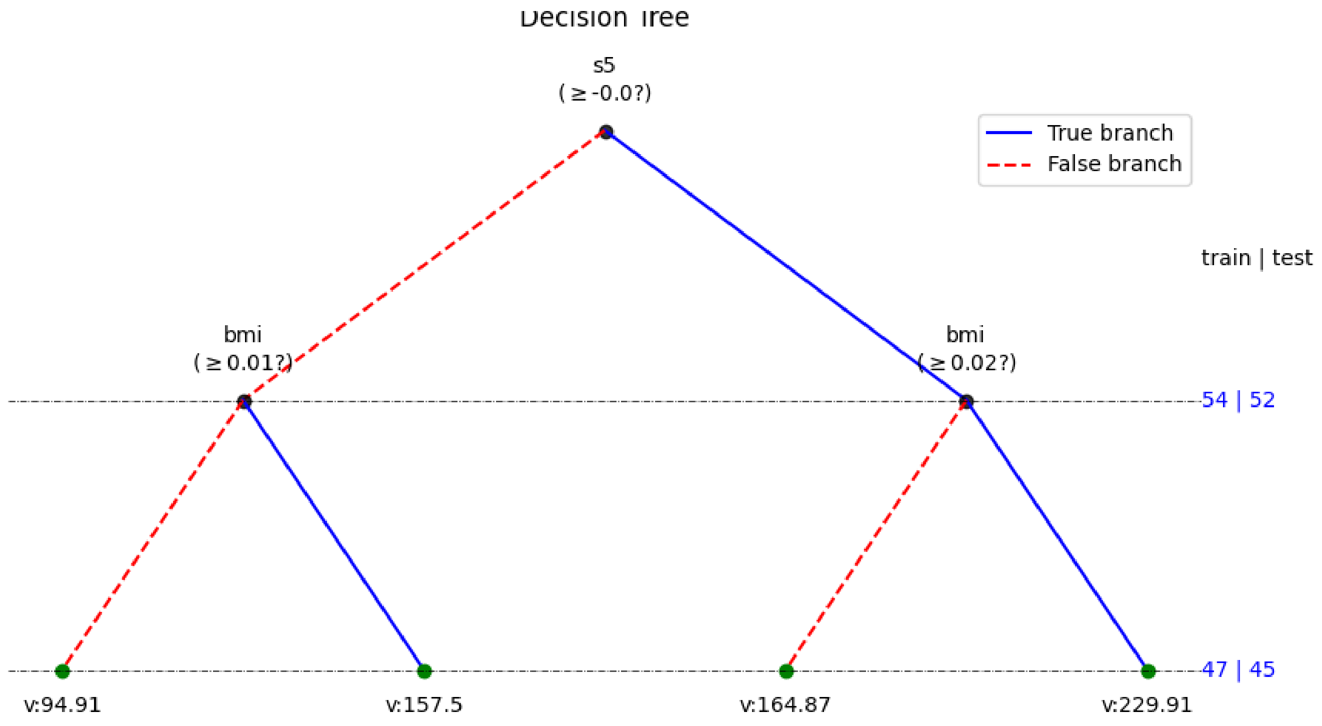
Decision Trees: Regression

- Example: Increasing the ***depth*** of tree, more flexibility and ***overfitting***



Decision Trees: Regression

- Example: Increasing the ***depth*** of tree, more flexibility and ***overfitting***



Avoid overfitting:

- Stopping criteria,
- Small depth
- Shrink/prune (use spkit)

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